

Automated MLOps-Driven YOLO Framework for Drone-Based Plant Disease Detection

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Abstract. This paper introduces a novel drone-based plant disease detection system optimized for efficient and scalable deployment using MLOps. Utilizing the CADI AI dataset for cashew crop disease classification, it includes automated workflows for iterative training, testing, and deployment across YOLO architectures (YOLOv5, YOLOv8, YOLOv9, and YOLOv10). Advanced data augmentation and incremental dataset expansion, growing from 757 training images to the full dataset, ensure fair evaluations and model optimization. YOLOv5 achieved a peak mAP@50 of 59.4%, followed by YOLOv8 with 50.1%. Iterative fine-tuning revealed YOLOv9's superior insect detection performance (mAP@50: 70.9 %) and YOLOv10's excellence in abiotic stress detection (mAP@50: 77.3%). This study highlights MLOps' role in real-time model deployment and benchmarking, showcasing robust object detection capabilities and emphasizing iterative optimization and auto-deployment strategies to address dataset imbalance and enhance precision agriculture.

Keywords: Crop Disease Detection, YOLOv5, YOLOv8, YOLOv10, YOLOv9, Object Detection

1 Introduction

The recent developments in computing technologies, such as machine learning and deep learning, have offered a huge transformation in agriculture by making plant disease detection automated, instead of manual and labor-intensive, with efficient data-driven systems. This important transformation is crucial to help hopefully satisfy the increasing global need for food security and optimization of crop yields. Traditional plant disease detection methods include field inspection manually, which wastes time and is exposed to human error; hence, it is not suitable for large-scale agricultural operations. These limitations put claims on scalable and efficient solutions towards modern precision agriculture.

Drone technology, mixed with advanced deep learning models, has come out as a quite transformative approach for real-time agricultural monitoring. Equipped with high-resolution cameras, drones have the ability to capture detailed imagery, allowing early disease detection and timely interventions to decrease the losses of crops [3]. However, applying computer vision models in agricultural environments offers unique challenges. Variability in lighting, weather,

and crop growth stages can hugely affect model accuracy and generalizability [7, 11]. Moreover, most of the already present systems lack scalability and automation, particularly in the iterative development and deployment of detection models. In order to surpass these present obstacles, this paper introduces

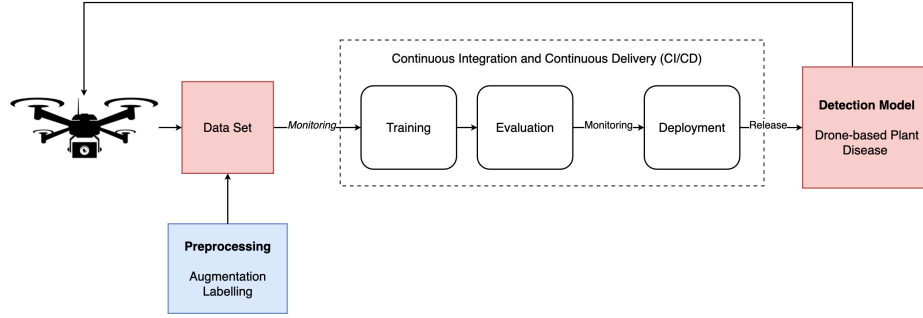


Fig. 1. Proposed automated MLOps pipeline for drone-based plant disease detection. The pipeline leverages the usage of continuous integration and continuous delivery (CI/CD) with iterative training, evaluation, and deployment of YOLO models, ensuring real-time UAV application integration and scalability.

an automated MLOps (Machine Learning Operations) pipeline designed to integrate continuous training, testing, and deployment of state-of-the-art YOLO object detection models (YOLOv5, YOLOv8, YOLOv9, and YOLOv10). The proposed pipeline, shown in Figure 1, allows drone-captured imagery and advanced data augmentation techniques to iteratively improve model performance. The pipeline starts with preprocessing, where images are augmented and labeled before being fed into the dataset module. It then uses continuous integration and continuous delivery (CI/CD) stages for training, evaluation, and deployment of YOLO models, backed by real-time monitoring. This iterative process ensures seamless scalability and model optimization for deployment in UAV-based applications.

The key to this approach is the iterative measuring of dataset sizes and the application of hyperparameter optimization to actually face the dataset imbalance and enhance model robustness under changing environmental conditions [4, 9]. By basically automating model training and deployment, the presented pipeline allows faster adaptation to new data and consistent improvements in detection accuracy, laying the foundations for real-world applications in precision agriculture.

2 Related Work

The detection and prediction of plant diseases has significant attention due to the increasing demand for efficient, scalable agricultural solutions. In a tradi-

tional sense, disease detection has mainly relied on actual visual inspection by agronomists, a method that is quite time-consuming, labor-intensive, and exposed to human faults. This has encouraged researchers to explore other options like automated approaches using machine learning (ML) and deep learning (DL). Convolutional neural networks (CNNs) have been especially significant in this domain, achieving remarkable results in image-based plant disease detection. For instance, CNN models have shown over 99% accuracy in classifying different plant diseases [2]. These advancements underscore CNNs' ability to handle large, complex datasets, showing a significant shift towards automated and reliable disease prediction in agriculture. However, recent literature reviews on DL in agriculture show that we are not quite there yet, particularly in the application of advanced visualization techniques and modifications to established models to further enhance plant disease identification [10].

Object detection algorithms have further propelled progress in plant disease detection, moving beyond traditional feature extraction methods that depend on color, texture, and shape. These regular techniques often struggle with generalization across actual different datasets and require vast manual engineering. In contrast, object detection deep learning-based models, such as YOLO (You Only Look Once), have altered the field dramatically by enabling real-time detection with competitive accuracy [6]. YOLO's speed and precision have shown promising results in agricultural settings, where rapid identification of diseases, pests, and anomalies is hugely important for timely intervention.

The iterative, data-driven nature of machine learning has created the need for new frameworks around model development management and deployment within complex environments. Unlike traditional systems, ML models require robust data management and versioning for reproducibility and experimentation. In the context of precision agriculture, this iterative process has allowed the continuous improvement of models tailored for specific datasets to ensure greater accuracy and scalability. Effective pipeline automation for model training, evaluation, and deployment supports this lifecycle, motivating researchers to reach optimized empowering performance for a range of challenges, such as plant disease detection.

The usage of drone technology with deep learning models has offered new opportunities for plant disease prediction at scale. Drones equipped with high-resolution RGB and multispectral cameras provide a unique vantage point, enabling the monitoring of huge agricultural areas. This approach offers higher spatial resolution and more frequent monitoring compared to traditional satellite or ground-based methods. Drones have proven particularly effective for early-stage disease detection, capturing small changes in crop health before the disease spreads. While satellite-based monitoring is valuable for large-scale assessments, it is limited by lower spatial resolution and revisit frequency, making it less effective for detecting early-stage diseases. Ground-based methods, though accurate, are labor-intensive and impractical for large-scale operations.

Fairly recent studies have shown the effectiveness of combining deep learning models with drone-captured imagery for plant disease detection. CNN-based

models have shown high accuracy in identifying plant diseases from drone imagery, even under challenging environmental conditions [8]. This harmony between drone technology and advanced machine learning techniques provides a scalable, efficient approach to crop health monitoring, allowing for timely interventions and optimized resource allocation. Consequently, drones paired with state-of-the-art object detection models are becoming essential tools in precision agriculture, enhancing disease prediction and prevention strategies [5].

3 Methodology

This paper presents a robust framework for detecting plant diseases in cashew orchards using drone-acquired imagery and deep YOLO models, namely, YOLOv5, YOLOv8, YOLOv9, and YOLOv10. The proposed framework integrates MLOps, which enables scalable and iterative improvements by automating model training, evaluation, deployment, and monitoring. This work uses the CADI-AI dataset to overpower challenges like dataset variability and environmental conditions through iterative training, advanced data augmentation, and automated workflows.

The MLOps of the study includes:

- **Dataset Management:** Automated data augmentation and annotation to handle dataset imbalance.
- **Continuous Monitoring:** Real-time tracking of training and evaluation metrics.
- **Incremental Scaling:** Iterative training with progressively larger datasets to enhance generalization.
- **Automated Deployment:** Real-time updates and deployment for adaptable YOLO models.

In this pipeline, shown in Figure 1, it integrates data collection, preprocessing, iterative training, and real-time deployment is integrated, with steps detailed in Algorithm 1. This methodology gives a scalable and efficient solution for drone-based plant disease detection, advancing precision agriculture.

3.1 Data Collection and Preprocessing

This particular pipeline we are using begins with the collection of high-resolution imagery (1600x1300 pixels) from drones equipped with P4 Multi-spectral cameras [1]. These images, annotated for disease categories such as insect damage, abiotic stress, and diseases, are preprocessed to create a structured dataset. Preprocessing steps involves:

1. **Data Augmentation:** Used techniques such as random cropping, flipping, and brightness adjustments simulate diverse environmental conditions and mitigate dataset imbalance.
2. **Labeling and Structuring:** Images are paired with annotations to create training, validation, and testing subsets. The preprocessed dataset is denoted as $\mathcal{D} = \{(\mathcal{I}, \mathcal{L})\}$, where $\mathcal{I}$ are images and $\mathcal{L}$ are labels.

3.2 Iterative Training with Continuous Integration

The core of the framework utilizes MLOps to implement iterative training with continuous integration (CI). The dataset is divided into six incremental stages, starting from 757 training images and gradually incorporating the full dataset. For each iteration:

1. **Model Selection:** A YOLO architecture (YOLOv5, YOLOv8, YOLOv9, or YOLOv10) is initialized with predefined hyperparameters.
2. **Training and Monitoring:** The model is trained on the current dataset subset $\mathcal{D}_{train}$ while performance metrics (e.g., mAP, precision, recall) are monitored in real-time.
3. **Hyperparameter Tuning:** Metrics from the validation subset $\mathcal{D}_{val}$ guide the adjustment of hyperparameters for improved model performance.

The training process and iterative improvements are outlined in Algorithm 1.

3.3 Evaluation and Continuous Delivery

Once trained, the model undergoes rigorous evaluation on $\mathcal{D}_{val}$ and $\mathcal{D}_{test}$ to ensure it meets predefined performance thresholds. The evaluation metrics include:

- **Mean Average Precision (mAP)** at IoU thresholds (mAP50, mAP50-95)
- **Precision (P)** and **Recall (R)**
- **Class-wise performance** for insect, abiotic stress, and disease detection

If the model achieves satisfactory results, it proceeds to deployment; otherwise, configurations are updated, and retraining is initiated.

3.4 Deployment and Real-Time Monitoring

The final YOLO model $\mathcal{M}_{final}$ is deployed for real-time inference on new drone imagery. The deployment pipeline includes:

1. **Integration with UAV Systems:** Models are deployed to inference engines compatible with drone systems, enabling real-time disease detection.
2. **Monitoring and Feedback:** Continuous monitoring of deployed model performance provides insights into real-world efficacy and guides periodic retraining with new data.

Algorithm 1 Automated MLOps Pipeline for Drone-Based Plant Disease Detection

 Drone-captured dataset $\mathcal{D}$ Deployed YOLO detection model $\mathcal{M}_{deployed}$

Step 1: Data Collection and Preprocessing**Input:** Images captured by drones.

1. Capture high-resolution images ($\mathcal{I}_{raw}$).
2. Perform data augmentation (e.g., flipping, cropping, brightness adjustment) to generate $\mathcal{I}_{augmented}$.
3. Annotate images and prepare labels ($\mathcal{L}$).
4. Store preprocessed dataset $\mathcal{D} = \{(\mathcal{I}_{augmented}, \mathcal{L})\}$.

Step 2: Training Process with Continuous Integration (CI)**Input:** Dataset $\mathcal{D}$, YOLO model configurations $\mathcal{C}$.

5. Initialize YOLO model $\mathcal{M}$ (e.g., YOLOv5, YOLOv8, YOLOv9, YOLOv10).
6. Split $\mathcal{D}$ into training ($\mathcal{D}_{train}$), validation ($\mathcal{D}_{val}$), and testing ($\mathcal{D}_{test}$) subsets.
7. Train $\mathcal{M}$ on $\mathcal{D}_{train}$ while monitoring performance metrics (e.g., mAP, precision, recall).
8. Evaluate $\mathcal{M}$ on $\mathcal{D}_{val}$ for hyperparameter tuning and performance tracking.

Step 3: Evaluation and Continuous Delivery (CD)**Input:** Trained YOLO model $\mathcal{M}$, validation results.

9. Validate $\mathcal{M}$ on $\mathcal{D}_{val}$ and ensure metrics meet the required threshold.
10. If performance is satisfactory, move $\mathcal{M}$ to the deployment stage.
11. Otherwise, update configurations $\mathcal{C}$ and retrain $\mathcal{M}$ (go to Step 5).

Step 4: Deployment and Real-Time Monitoring**Input:** Final YOLO model $\mathcal{M}_{final}$, testing dataset $\mathcal{D}_{test}$.

12. Deploy $\mathcal{M}_{final}$ for real-time inference on drone imagery.
 13. Continuously monitor $\mathcal{M}_{deployed}$ using inference metrics and feedback.
 14. Periodically update $\mathcal{M}_{deployed}$ with new data from the pipeline.
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4 Experimental Setup

The experimental setup was designed to evaluate the proposed MLOps-integrated pipeline for drone-based plant disease detection, utilizing the CADI-AI dataset. This high-resolution dataset contains 4,736 drone-acquired images of cashew crops annotated in YOLO format, categorized into three classes: abiotic stress, insect damage, and disease. The dataset was partitioned into 3,788 images for training, 710 for validation, and 238 for testing, ensuring a comprehensive evaluation of the models.

Iterative Training Process To enhance scalability and generalization, an **iterative training approach** was employed across five stages:

- **Iteration 1:** Initial training on 757 images.
- **Iteration 2:** Dataset expanded to 1,515 images by adding 757 more.

- **Iteration 3:** Training on 2,272 images.
- **Iteration 4:** Dataset scaled to 3,030 images.
- **Iteration 5 (Final):** Complete dataset of 3,788 images used with extensive augmentation and class balancing.

This progressive strategy allowed the models to learn incrementally, addressing class imbalances and improving detection robustness.

Data Augmentation and Generalization To simulate real-world environmental variability and improve generalization, various augmentation techniques were uniformly applied across all iterations:

- Horizontal and vertical flipping
- Brightness and contrast adjustments
- Random gamma modifications
- Gaussian blur
- Geometric transformations (scaling, shifting, and rotation)

These augmentations enhanced the model’s capacity to adapt to unseen data, particularly under challenging environmental conditions.

MLOps Integration for Automated Deployment The integration of **MLOps (Machine Learning Operations)** streamlined the model lifecycle by automating key processes:

- **Automated Deployment:** Continuous integration and deployment pipelines ensured real-time availability of high-performing models.
- **Automated Testing:** Each iteration underwent rigorous evaluation using metrics such as precision, recall, **mAP50**, and **mAP50-95**.
- **Hyperparameter Tuning:** Automated fine-tuning of learning rates, batch sizes, and other parameters accelerated convergence and improved performance.

Training Configuration Training was done using the **Adam optimizer** with an initial learning rate of 0.001. A **cosine annealing scheduler** dynamically reduced the learning rate to ensure smooth convergence.

Hardware and Environment All experiments were conducted on **Google Colab** using an **NVIDIA Tesla P100 GPU with 16 GB VRAM**. The iterative training process was applied across YOLOv5, YOLOv8, YOLOv9, and YOLOv10 to ensure fair and consistent comparisons across architectures.

5 Results and Discussion

This section describes the performance evaluation of YOLOv5, YOLOv8, YOLOv9, and YOLOv10 based on the metrics of mAP50, mAP50-95, precision, and recall for the CADI-AI dataset. In particular, models were tested in detecting three

Table 1. Performance Metrics of YOLO Models (M) Across Iterations (I). The table highlights improvements in mAP50, P, and R for different YOLO versions, with key results emphasized.

M	I	mAP50	mAP50-95	P	R	Ins	mAP50	Abi	mAP50	Dis	mAP50
YOLOv5	1	19.3	7.59	43.5	19.3	28.5		5.5		14.2	
	2	31.8	12.4	54.1	33.0	46.5		23.4		25.4	
	3	47.8	18.7	66.3	46.5	74.9		27.8		40.6	
	4	50.6	20.4	66.4	50.8	75.4		29.3		47.2	
	5	59.4	23.2	68.0	59.5	75.1		52.2		50.9	
YOLOv8	1	33.5	11.9	47.3	37.5	20.6		52.5		27.4	
	2	40.5	14.8	52.2	43.1	23.2		66.5		31.9	
	3	45.1	17.2	58.9	45.7	26.6		72.0		36.9	
	4	45.5	17.8	58.0	46.0	26.7		71.5		38.2	
	5	50.1	19.8	63.7	47.1	26.8		75.3		42.1	
YOLOv9	1	20.3	8.01	38.9	21.0	29.7		17.1		14.0	
	2	34.0	13.4	54.0	33.4	50.1		25.1		26.8	
	3	48.6	20.2	62.0	46.9	70.9		25.7		44.0	
	4	50.5	21.6	64.1	47.4	74.1		31.9		45.6	
	5	60.1	25.2	64.7	57.5	76.6		51.7		50.4	
YOLOv10	1	31.3	12.0	44.4	35.1	24.0		48.3		21.5	
	2	42.7	16.6	58.0	40.9	25.3		67.4		35.3	
	3	45.2	17.5	61.6	43.6	27.1		70.5		38.0	
	4	45.4	18.2	61.8	44.1	26.3		69.8		47.6	
	5	57.1	23.0	67.6	55.8	46.4		77.3		47.6	

categories of stresses, namely, insect damage, abiotic stress, and diseases. The MLOps pipeline allowed the iterative training strategy that led to systematic improvements in the performance, summarized in Table 1 and Table 2.

The results demonstrate consistent improvements across YOLOv5 iterations, starting with a mAP50 of 19.3% in Iteration 1 and achieving 59.4% in Iteration 5. The model excelled in detecting insect damage, reaching a mAP50 of 75.1% in the final iteration, while achieving moderate performance in abiotic stress (mAP50: 52.2%) and disease detection (mAP50: 50.9%). These improvements highlight the importance of advanced data augmentation and hyperparameter tuning in addressing dataset variability.

YOLOv8 showcased its strength in abiotic stress detection, achieving a peak mAP50 of 75.3% in Iteration 5. However, its overall performance (mAP50: 50.1%) and insect detection capabilities (mAP50: 26.8%) lagged behind YOLOv5 and YOLOv9. This indicates that YOLOv8’s architecture is particularly suited for identifying subtle patterns, such as those in abiotic stress.

YOLOv9 emerged as the most balanced model, achieving the highest overall mAP50 of 60.1% in Iteration 5. It demonstrated exceptional insect detection performance (mAP50: 76.6%) and competitive results in abiotic stress (mAP50: 51.7%) and disease detection (mAP50: 50.4%). These outcomes underscore YOLOv9’s

Table 2. Experimental Procedures Using YOLO Architectures Across Iterations.

I	Ratio (%)	Images	YOLOv5	YOLOv8	YOLOv9	YOLOv10
1	20	757	19.3	33.5	20.3	31.3
2	40	1515	31.8	40.5	34.6	42.7
3	60	2272	47.8	45.5	48.6	45.2
4	80	3030	50.6	45.5	50.5	45.4
5	100	3788	59.4	50.1	60.1	57.1

ability to generalize effectively across stress categories, attributed to its enhanced gradient flow and feature aggregation strategies.

YOLOv10 excelled in abiotic stress detection, achieving the highest mAP50 of 77.3% in this category during Iteration 5. Although its overall mAP50 of 57.1% and recall of 55.8% were slightly lower than YOLOv9, YOLOv10 demonstrated improved performance in high-resolution imagery scenarios. However, its insect detection (mAP50: 46.4%) and disease detection (mAP50: 47.6%) results were comparatively lower, likely due to architectural trade-offs.

As shown in Table 2, the iterative training strategy expanded the dataset size from 20% to 100% in five steps. This resulted in consistent improvement in performance with the two systems: YOLOv9 and YOLOv10 significantly improved. For example, YOLOv9 improved from 20.3% mAP50 in Iteration 1 to 60.1% mAP50 in Iteration 5, while YOLOv10 similarly improved from 31.3% to 57.1% mAP50. These findings demonstrate that MLOps could be an effective pipeline that will enable an orderly automation process in training, testing, and deployment towards the better performance optimization of both.

6 Conclusion

This work will demonstrate how some of the most recent advanced object detection models, namely YOLOv5, YOLOv8, YOLOv9, and YOLOv10, perform in general object detection based on drone-acquired imagery. To achieve better accuracy, scalability, and deployability in real time, iterative model training was performed using an MLOps pipeline. The best overall general object detection result was given by YOLOv9, while abiotic stress could be better identified with YOLOv10-the strength of one architecture over another. Both YOLOv8 and YOLOv5 demonstrated progressive improvements that substantiate the benefits derived from iterative dataset augmentation and automated optimization. However, the challenges remain in light of the environmental variability and subtleness of visual features related to abiotic stress and disease detection. Future efforts should go more into diversification of datasets and integration of multi-spectral or hyperspectral imaging in order to raise the detection accuracy of subtle stress factors. This can enable real-time deployment during drone operations, supported by iterative MLOps pipelines, to revolutionize precision agriculture through timely disease detection and intervention that improve crop health and yield.

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